

Chapter 3

System Design

Learning Outcomes

Upon completion of this chapter students will know

- ❖ How to use multiplexed bus and signals in 8086
- ❖ How to separate the addresses from data and status signals
- ❖ Memory organization of 8086 based systems
- ❖ Interfacing memory and input/output peripheral devices to 8086
- ❖ Memory mapping techniques
- ❖ 8086 based system design

3.1 General Concepts of System Design

A microprocessor based system comprises of both hardware and software. The hardware is embedded in a board with other peripheral onboard resources interfaced to the microprocessor. The onboard resources like memory (ROM, RAM), I/O devices, Timers, etc. should be connected in a structured manner so that accessing one element does not conflict with others. The following points must be kept in mind while designing a hardware microprocessor board.

- ❖ All address lines out of the microprocessor should be latched to the board address bus by external latches.
- ❖ The external address (A0-A19) bus is connected in parallel to all resources in the board.
- ❖ The external data bus (D0-D15) is connected in parallel to all resources in the board.
- ❖ Data bus of any device should be kept in tri-state (high impedance) mode, and be connected to the external data bus only when requested from the processor.

3.2 8086 Interface to Memory

In a microprocessor system, the machine code instructions in hex code are stored in Read only Memory (ROM) and variables are stored in random access memory (RAM). Receiving (reading) program codes from the ROM is called instruction fetching. In the process of computing the processor may need to send (write) data to memory for temporary storage.

During these read/write operations, the processor should enable the intended device only where the read/write operation will be performed. No two devices should be allowed to receive or send data/code through the data bus.

Since 8086 is a complex processor supporting complex and variable length instructions, it may need to access 8-bit or 16-bit data depending on the instruction.

8086 and all its derivatives divide the total accessible memory into two distinct parts, even and odd as shown in Fig. 3.1. The even and odd memory locations must reside in separate devices. With the 20-bit external address bus, 8086 can address up to 1

MB space. However, for a particular program, it can address effectively a maximum of 64-kB from a base hold by the segment registers.

As per Intel convention, lower memory locations hold lower byte. The lower memory locations by default are the even addresses. If a word is stored in memory, it is implied that the low byte be stored in even location and the high byte stored in odd location. Note that if words are stored from odd locations, the performance of the processor slows down as it takes two machine cycles for accessing the data. For addressing even and odd locations separately, 8086 provides the BHE# pin. Conventionally, a device containing even addresses are enabled using A0 and odd addressed devices are enabled using BHE#. Fig. 3.2 shows a typical arrangement of addressing even and odd locations using the A0 and BHE# signals.

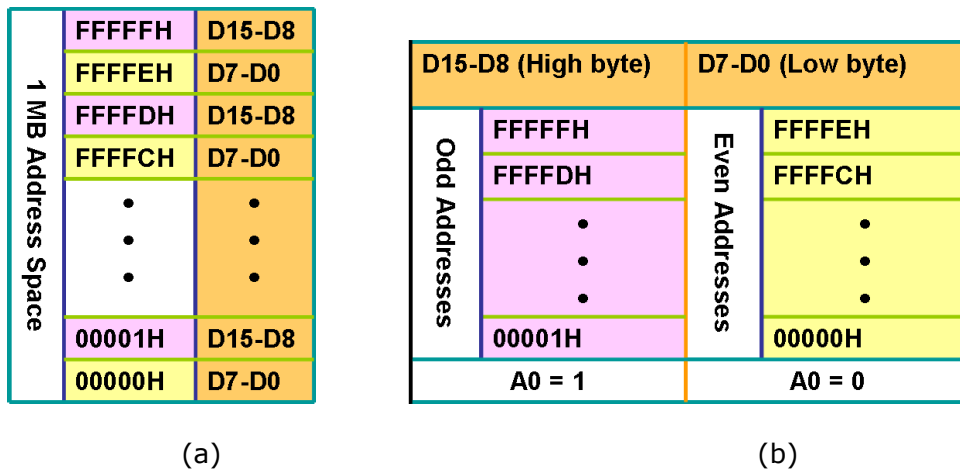


Fig. 3.1 Addressing memory space of 8086, (a) total 1 MB memory, (b) memory divided into even and odd.

3.3 Memory Interfacing to 8086

A microprocessor system requires both program memory and data memory. Memory Interfacing to 8086

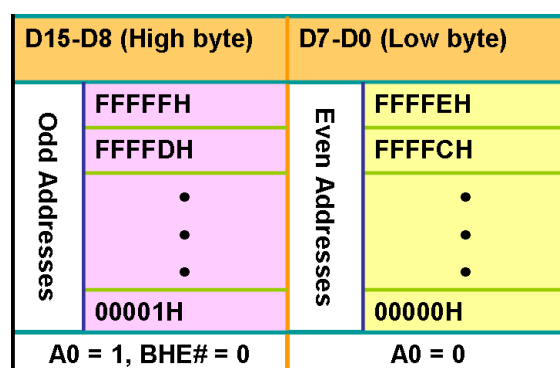


Fig. 3.2 Accessing even and odd memory location using BHE# and A0 signals.

The instruction codes for the program are stored in program memory while the data required for the program are stored in data memory. Program memory is usually stored in ROM (PROM, EPROM, EEPROM, FLASH etc.) and the data memory is stored in RAM. For a general purpose application like a PC, however, the user may load and run program from RAM.

ROM chips can be read only, and usually have one signal pin for chip enable (CE) and another for output enable (OE). The chip enable signal activates the chip and the output enable signal connects the internal data bus to the output data pins.

RAM chips can be both read and write, hence have one more pin called write enable (WE) in addition to CS and OE.

The addressing rule is that the master chip enable (CE) should be selected using the **unused** address lines, A0 or BHE. The OE or WE must be addressed using the CS, RD and WR signals.

3.3.1 Interfacing 27 series EPROM's

The 27 series EPROM's are designed to be compatible with 8086 that can be operated in **Read Mode** and **Standby Mode**. Consider 2764 having 8k-byte EPROM as shown in Fig. 3.3.

Read Mode

The 2764 has two control functions, both of which must be logically satisfied in order to obtain data at the outputs. Chip Enable (CE) is the power control and should be used for device selection. Output Enable (OE) is the output control and should be used to gate data to the output pins, independent of device selection. Assuming that the addresses are stable data is available at the outputs after the falling edge of OE, assuming that CE has been low.

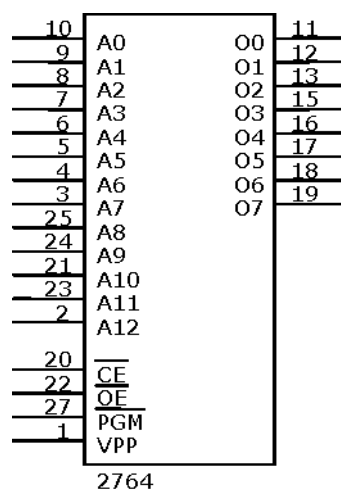


Fig. 3.3 Pin configuration of 2764 EPROM.

Standby Mode

The 2764 has a standby mode which reduces the maximum active power current. The 2764 is placed in the standby mode by applying a TTL high signal to the CE input. When in the standby mode, the outputs are in a high impedance state, independent of the OE input.

Examples of interfacing ROM, RAM and I/O device to an 8086 system are discussed in the following sections.

3.4 8086 System with 4k ROM and 4k RAM

Let us consider a system containing 4k byte ROM and 4k byte RAM and 8-bit input/output. We can use two 2716 for ROM and two 6116 for RAM, and two 74LS373 latches for input/output. The ROM needs addressing for reading only, while, the RAM needs both read and write. For addressing a ROM we need RD#, BHE# and A0. For addressing RAM we need RD#, WR#, BHE# and A0. The memory space for ROM and RAM are shown in Table 3.1.

Table 3.1 Memory mapping for 8086 system with 4k-byte ROM and 4k-byte RAM.

	5 th HEX DIGIT				4 th HEX DIGIT				3 rd HEX DIGIT				2 nd HEX DIGIT				1 st HEX DIGIT			
Device	A19	A18	A17	A16	A15	A14	A13	A12	A11	A10	A9	A8	A7	A6	A5	A4	A3	A2	A1	A0
2716 (EVEN)	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0
	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑
	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
2716 (ODD)	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0
	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑
	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
UNUSED MEMORY SPACE																				
6116 (EVEN)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑
	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
6116 (ODD)	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	0
	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑
	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1
	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑	↑
	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓

The program memory (ROM) addresses are on the top from FF000H to FFFFFH. The RAM addresses are on the bottom occupying spaces 00000H to 00FFFH.

3.4.1 Memory (ROM and RAM) Addressing

The next step is the memory decoding that can be done according to Table 3.2. From Table 3.1 and Table 3.2, it is evident that the address lines A12-A19 are unused and hence can be used for master chip select. The simplest case will be to use A12, and leaving A13-A19 unconnected. This will result in memory fold back as A12 changes state from 0 to 1 on every 8k byte step. This will not pose any problem for the 8086 system as no devices are placed in the remaining memory space.

Table 3.2 Memory decoding for 4k byte ROM and 4k byte RAM

A19	A18	A17	A16	A15	A14	A13	A12	BHE#	A0	RD#	WR#	Device to be selected
1	1	1	1	1	1	1	1	1	0	0	1	EVEN ROM (2716-1)
1	1	1	1	1	1	1	1	0	1	0	1	ODD ROM (2716-2)
0	0	0	0	0	0	0	0	1	0	0	1	READ EVEN RAM (6116-1)
0	0	0	0	0	0	0	0	1	0	1	0	WRITE EVEN RAM (6116-1)
0	0	0	0	0	0	0	0	0	1	0	1	READ ODD RAM (6116-2)
0	0	0	0	0	0	0	0	0	1	1	0	WRITE ODD RAM (6116-2)

As seen from Fig. 3.4, A12 is inverted and OR'ed with A0 to select the even ROM chip 2716-1. The output enable (OE#) of 2716-1 is selected OR'ing the CE# and

RD# signals 2716-1. The selection of the even ROM, 2716-2 is the same as 2716-1 with A0 replaced by BHE#.

The master chip select of the even RAM (6116-1) is enabled using A12 and A0 OR'ed together. The output enable (OE#) is selected OR'ing the master chip enable (CE#) and RD# pins. The write enable (WE#) is selected OR'ing the CE# and WR# pins. Selection of the odd RAM (6116-2) is same as 6116-1 with BHE# replacing A0.

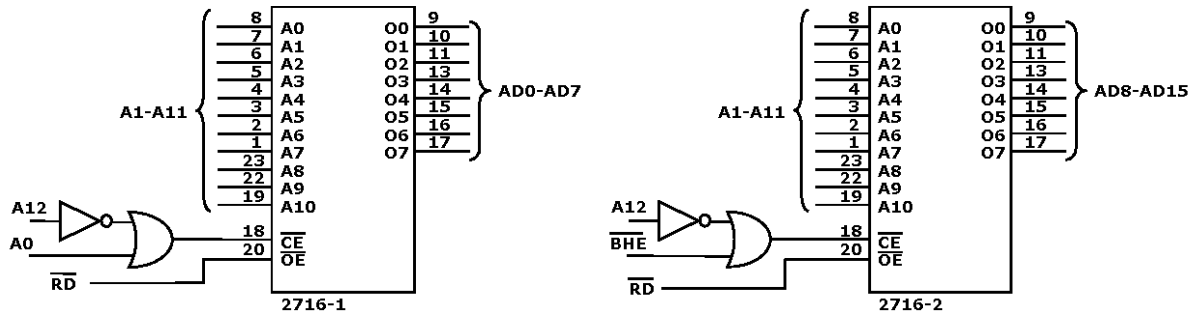


Fig. 3.4 Interfacing 4k byte EPROMs to 8086, the address space is FF000H – FFFFFH.

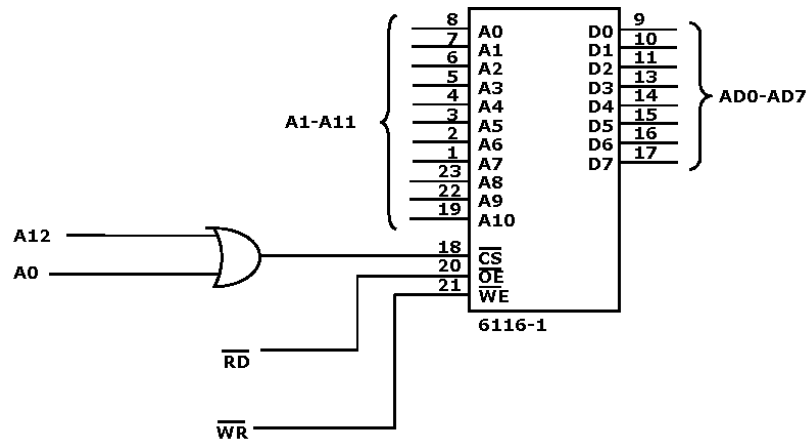


Fig. 3.5 Interfacing 2k byte (even addresses) RAM to 8086, the address space is 00000H – 00FFEH.

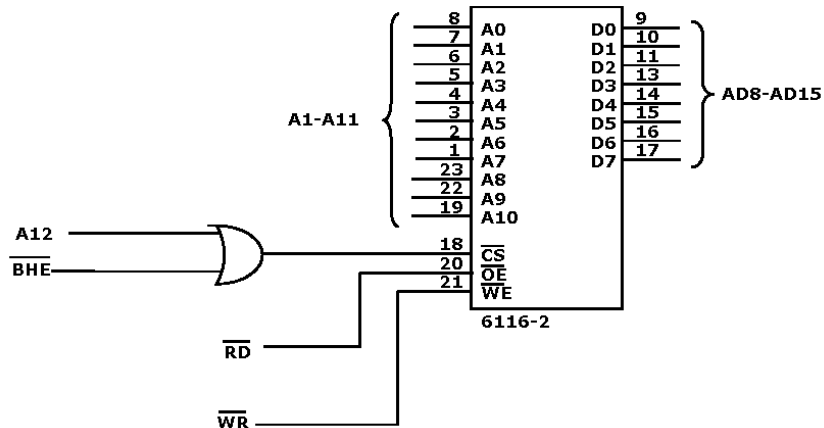


Fig. 3.6 Interfacing 2k byte (odd addresses) RAM to 8086, the address space is 00001H – 00FFFH.

System design, however, without memory fold back is also possible where whole memory segment has to be selected using a ROM decoder. ROM and RAM chip selects eliminating fold-back are shown in Fig. 3.8 and Fig. 3.8.

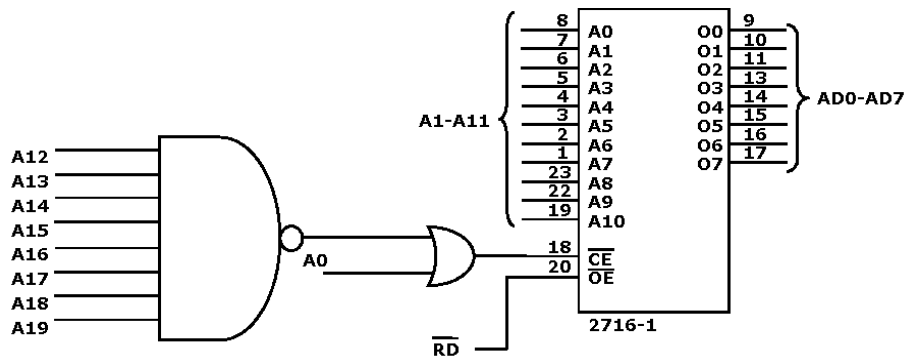


Fig. 3.7 Circuitry for addressing 4k-byte ROM without memory fold-back using full memory decoding technique.

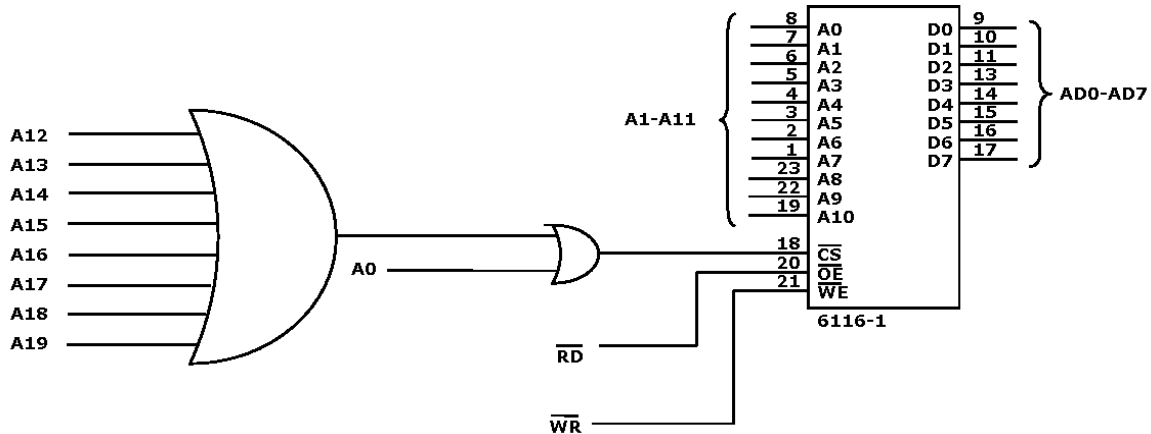


Fig. 3.8 Circuitry for addressing of 4k-byte RAM without memory fold-back using full memory decoding technique.

3.4.2 Simple I/O for 8-bit Input and 8-bit Output

There are standard devices like 8255 for parallel I/O applications. Here we use a simple device for input and output operations. Two 74LS373 will be used for this purpose, one for 8-bit input and the other for 8-bit output. Note that 8086 provides separate instructions (IN and OUT) for accessing I/O devices. During these IN and OUT instructions, 8086 outputs LOW on its M/I0# pin. Hence for accessing I/O devices, the M/I0# pin can be used along with other address lines.

Although one device acts as input and the other as output, it is possible they both have the same address. The address is 00H, the input port is activated when RD# is LOW and the output port is activated when WR# is LOW.

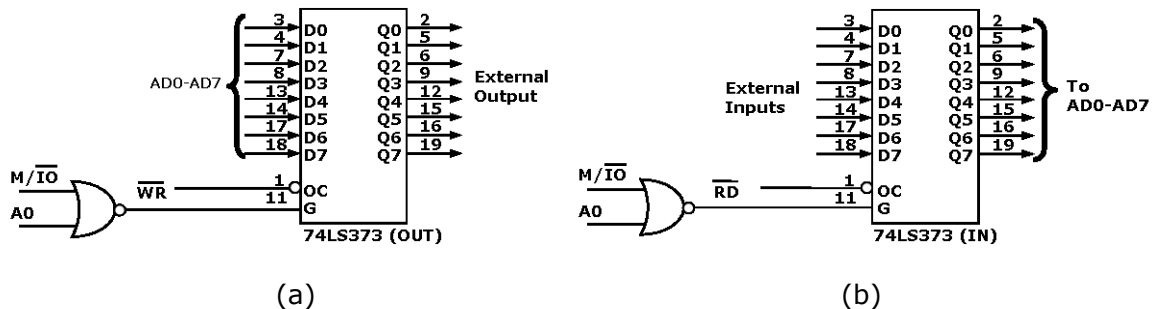


Fig. 3.9 Interfacing 8-bit I/O, a) input and b) output, implemented using two 74LS373 having address 00H.

Review Questions

1. Explain the mechanism of accessing memory in 8086.
2. How does 8086 addresses even and odd memories?
3. Explain with a suitable diagram the selection and addressing read only memories (ROM) with 8086.
4. Two EPROM's each 32k-bytes are to be addressed in even and odd locations from FFFFH:0000H. Give detail connections for chip enable, read and memory addressing.
5. Explain with a suitable diagram the selection and addressing random access memories (RAM) with 8086.
6. Two static RAMs each 32k-bytes are to be addressed in even and odd locations from 0000H:0000H. Give detail connections for chip enable, read, write and memory addressing.
7. How does 8086 addresses I/O devices?
8. Briefly explain with necessary schematics the addressing of I/O devices in odd and even locations, considering 8-bit address.
9. Briefly explain with necessary schematics the addressing of I/O devices in odd and even locations, considering 16-bit address.
10. An I/O device with an 8-bit data bus has four accessible registers, with selection lines L0 and L1. If the addresses of the registers are 80H, 82H, 84H and 86H respectively, develop an address map and hence show the connection circuitry.